RESULTS CHAPTER DRAFT · TOMATO GROWTH-STAGE RECOGNITION

Segmentation-Guided Growth-Stage Classification: Full Results Summary

Consolidated results for the ResNet50 background-dependency study, the DeepLabV3-ResNet50 segmentation pipeline, and the U-Net comparison pipeline — accuracy, precision, recall, F1, and IoU across every experiment, with the practical issues encountered along the way.

Task 4-class tomato growth-stage classification (seeding · developing · flowering · fruiting) **Held-out test set** 160 images

Compiled 2 Aug 2026

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1. Experimental pipeline & dataset

All experiments classify tomato plant photographs into four growth stages — **seeding, developing, flowering, fruiting** — using a fine-tuned ResNet50. The central research question is whether a plain classifier trained on full photographs is actually reading the plant, or is partly reading the greenhouse/pot background around it, and whether a segment-then-classify pipeline (mask the plant out, then classify only that region) is both feasible with predicted masks and beneficial for accuracy. That gives three linked experiments, evaluated on the same 160-image held-out test set:

1. Classify (original photo) → 2. Classify (oracle-masked photo) → 3. Segment → predicted mask → classify

Stage 3 is run twice, once with a **DeepLabV3-ResNet50** segmenter and once with a **U-Net** segmenter, so the two architectures can be compared as drop-in replacements in an otherwise identical pipeline. Class support in the test set is imbalanced: developing 41, flowering 60, fruiting 21, seeding 38 — fruiting is the smallest class throughout and, as the results below show, consistently the hardest.

2. ResNet50 classification & background dependency

2.1 Baseline classifier — original (unmasked) images

ImageNet-pretrained ResNet50, fully fine-tuned (no frozen layers, 23.5M trainable parameters), `fc` replaced with a 4-way linear head. Adam, lr 1e-4, `ReduceLROnPlateau`, batch 32, 20 epochs, with random-resized-crop / flip / rotation / color-jitter augmentation. Trained on 687 images, validated on 135, tested on 160.

93.13%	0.93	0.91	0.92
TEST ACCURACY	MACRO PRECISION	MACRO RECALL	MACRO F1

Per-class test metrics — baseline (original-image) classifier

CLASS	SUPPORT	PRECISION	RECALL	F1
developing	41	0.93	1.00	0.96
flowering	60	0.90	0.92	0.91
fruiting	21	0.88	0.71	0.79
seeding	38	1.00	1.00	1.00
macro avg	160	**0.93**	**0.91**	**0.92**
weighted avg	160	0.93	0.93	0.93

Confusion matrix — rows = true class, columns = predicted class

	DEVELOPING	FLOWERING	FRUITING	SEEDING
developing	**41**	0	0	0
flowering	3	**55**	2	0
fruiting	0	6	**15**	0
seeding	0	0	0	**38**

Fruiting is the only weak point: 6 of 21 fruiting images are misread as flowering, dragging recall down to 0.71 while every other class is at or near ceiling.

2.2 Masked classifier — ground-truth background removed

Identical architecture and training recipe, retrained from ImageNet weights on the same photos with everything outside the ground-truth COCO polygon blacked out (684 train / 135 val / 160 test). This model is reused by both segmentation pipelines in §3–4.

91.87%	0.91	0.87	0.89
TEST ACC (MASKED→MASKED)	MACRO PRECISION	MACRO RECALL	MACRO F1

Per-class test metrics — masked classifier, evaluated on masked images

CLASS	SUPPORT	PRECISION	RECALL	F1
developing	41	0.98	0.98	0.98
flowering	60	0.88	0.95	0.91
fruiting	21	0.86	0.57	0.69
seeding	38	0.95	1.00	0.97
macro avg	160	**0.91**	**0.87**	**0.89**
weighted avg	160	0.92	0.92	0.91

Fruiting recall drops further here (0.57) — with the background gone, 8 of 21 fruiting images are confused with flowering.

2.3 Four-way cross-evaluation: does the model look at the plant, or the background?

Each of the two classifiers above is evaluated on *both* image types, producing a 2×2 grid that isolates how much of each model's accuracy depends on background context it was never supposed to need.

Train condition × eval condition — test accuracy

TRAINED ON	EVALUATED ON	TEST ACCURACY
original	original	**96.88%**
original	masked	76.88%
masked	original	35.63%
masked	masked	**91.87%**

READING THE DIAGONAL DROPS

Original-trained model, background stripped at test time: accuracy falls 96.88% → 76.88% (−20.0 pts). Fruiting precision and recall both collapse to **0.00** — the model never predicts "fruiting" once the background is removed, meaning part of how it recognised fruiting stage came from context, not the plant itself.

Masked-trained model, evaluated on real photos: accuracy collapses 91.87% → 35.63% (−56.25 pts). Flowering and seeding precision/recall both fall to **0.00** — the model only ever outputs "developing" or "fruiting," having learned features tied to the artificial black background and mask edges that do not exist in real deployment photos.

2.4 Grad-CAM background-attention analysis

To test the background-reliance finding independently of accuracy, Grad-CAM heatmaps were computed for both models on all 160 original test images, and the fraction of total activation energy falling *outside* the ground-truth plant polygon was measured (the "background attention fraction").

Mean background attention fraction (± std. dev.), n = 160

MODEL	BACKGROUND ATTENTION FRACTION
Original-trained	0.7105 ± 0.1351
Masked-trained	0.7299 ± 0.1465

TENSION WORTH REPORTING, NOT SMOOTHING OVER

Both models place **over 70% of their Grad-CAM activation mass outside the plant polygon**, and the masked-trained model's fraction is marginally *higher*, not lower — the opposite of what the accuracy-drop finding in §2.3 would predict. The two std. deviations overlap heavily, so the difference is not significant either way. This means the accuracy-based conclusion ("masked training reduces background reliance") is not corroborated by this particular Grad-CAM metric: both models appear to attend broadly to pot/greenhouse structure around the canopy regardless of training condition, even though only the original-trained model's accuracy actually degrades when that context is removed. Qualitatively, example heatmaps for both models sit mainly on the plant canopy with a halo bleeding onto the surrounding pot and background, without one model showing a starkly tighter focus than the other.

3. DeepLabV3-ResNet50 segmentation pipeline

3.1 Training setup

COCO-pretrained `deeplabv3_resnet50`, fine-tuned as a binary (plant vs. background) segmenter — classifier and auxiliary heads replaced with 1×1 convs to 2 classes (41.99M trainable parameters). Cross-entropy loss with the auxiliary head weighted at 0.4. Adam, lr 1e-4, `ReduceLROnPlateau`, batch 16, 15 epochs, joint image+mask augmentation (crop, flip, rotation, color jitter). Trained on 684 images, validated on 135, tested on 160 — the same split as the classifiers above. Training converged smoothly with no train/val divergence: training mean IoU rose from 0.581 (epoch 1) to 0.784 (epoch 15), validation IoU from 0.656 to a best of **0.7956**, in 11m 42s.

3.2 Segmentation quality (held-out test set)

0.7523	0.7565	90.10%	0.8542
MEAN IOU	FOREGROUND (PLANT) IOU	PIXEL ACCURACY	MACRO F1

Pixel-level precision / recall / F1 by class, test set

CLASS	PRECISION	RECALL	F1
Background	0.9450	0.9289	0.9368
Plant (foreground)	0.7482	0.7963	0.7715
Macro avg	**0.8466**	**0.8626**	**0.8542**

3.3 End-to-end pipeline (predicted mask → masked classifier)

This is the realistic, deployable number: the segmenter predicts a mask on a brand-new photo with no annotations, that mask blacks out the background, and the masked classifier from §2.2 classifies the result.

87.50%	0.85	0.89	0.86
PIPELINE ACCURACY	MACRO PRECISION	MACRO RECALL	MACRO F1

Per-class pipeline metrics, test set (n=160)

CLASS	SUPPORT	PRECISION	RECALL	F1
developing	41	0.95	1.00	0.98
flowering	60	0.92	0.73	0.81
fruiting	21	0.55	0.81	0.65
seeding	38	1.00	1.00	1.00
macro avg	160	**0.85**	**0.89**	**0.86**
weighted avg	160	0.90	0.88	0.88

A later standalone re-evaluation notebook recomputes this same pipeline from the saved checkpoints and reports 84.38% accuracy / 0.7507 mean IoU against an "originally reported" 85.00% / 0.7570 — a small (≤0.7 pt) discrepancy the source notebooks flag

inline but do not diagnose, most likely run-to-run non-determinism in mask thresholding or batch ordering. Both figures are reported here for transparency; the confusion matrix below is from the recomputed run.

Confusion matrix — recomputed run (rows = true, columns = predicted)

	DEVELOPING	FLOWERING	FRUITING	SEEDING
developing	**36**	5	0	0
flowering	1	**46**	13	0
fruiting	0	6	**15**	0
seeding	0	0	0	**38**

3.4 Oracle mask vs. predicted mask — the realism gap

> **WHAT PREDICTING THE MASK ACTUALLY COSTS**
>
> Feeding the masked classifier *ground-truth* polygon masks (§2.2) gives 91.87% accuracy; feeding it masks *predicted* by DeepLabV3 gives 87.50%. The **4.37-point gap** is the realistic cost of not having annotations at inference time — everything above that line requires oracle masks that don't exist for new, unlabelled photos.

3.5 5-fold cross-validation

Two separate CV studies were run on the pooled 979-image annotated set (`StratifiedKFold` by growth stage). The first retrains only the segmenter per fold and reuses the fixed classifier; the second retrains *both* the segmenter and the classifier per fold and chains them — the more realistic, if noisier, estimate.

Segmentation-only CV (classifier held fixed) — mean ± std across 5 folds

METRIC	MEAN	STD. DEV.
Test IoU	0.7466	±0.0131
Test pixel accuracy	0.9119	±0.0050
Precision (macro)	0.8527	±0.0049
Recall (macro)	0.8449	±0.0154
F1 (macro)	0.8485	±0.0086

Full end-to-end pipeline CV (segmenter + classifier both retrained per fold) — mean ± std

METRIC	SEGMENTATION HALF	PIPELINE (4-CLASS) HALF
IoU / Accuracy	0.7489 ± 0.0037	0.8223 ± 0.0216
Precision (macro)	0.8544 ± 0.0110	0.7712 ± 0.0674
Recall (macro)	0.8385 ± 0.0108	0.7580 ± 0.0527
F1 (macro)	0.8457 ± 0.0018	0.7408 ± 0.0430

The single fixed-split pipeline accuracy (87.50%) sits about 5.3 points above the 5-fold CV mean (82.23%) — the fixed test split happened to be somewhat favorable, and 82.23% ± 2.16 pts is the more defensible, variance-aware estimate of real-world pipeline accuracy to quote as the headline number.

4. U-Net segmentation pipeline (comparison)

4.1 Architecture and the crop-alignment problem

A U-Net (Ronneberger et al. design, trained from scratch — no pretrained backbone, 7.77M parameters, encoder channels 32→64→128→256→512, symmetric decoder with skip connections) trained in a separate project was loaded via its saved checkpoint and dropped into the identical segment→mask→classify pipeline used for DeepLabV3, so the two architectures could be compared on the same 160-image test set.

Doing this exposed a real geometry bug: U-Net's own training/inference convention is a *direct* resize to 224×224 (distorting aspect ratio), while DeepLabV3 and the classifier both use a `Resize(256) → CenterCrop(224)` convention. Applying a mask predicted in one geometry to an image built in the other silently misaligns mask and plant. Two variants were therefore evaluated:

- **U-Net (native resize):** U-Net's own direct-resize frame used for both mask prediction and the image handed to the classifier — internally consistent, but different preprocessing than the classifier was trained/evaluated with elsewhere.
- **U-Net (crop-aligned):** U-Net fed the same `Resize(256) → CenterCrop(224)` image DeepLabV3 uses, removing the alignment mismatch entirely — at the risk of showing U-Net a framing it never saw in training.

4.2 Segmentation quality — all three variants

Segmentation quality, test set (n=160)

SEGMENTER	FOREGROUND IOU	FOREGROUND DICE	MEAN IOU (BOTH CLASSES)
DeepLabV3-ResNet50	0.6230	0.7565	**0.7507**
U-Net (native resize)	0.6646	0.7844	**0.7926**
U-Net (crop-aligned)	**0.6674**	0.7827	0.7753

Foreground IoU by growth-stage class — where each segmenter struggles

CLASS	DEEPLABV3	U-NET (NATIVE)	U-NET (CROP-ALIGNED)
developing	0.6883	0.6865	0.6730
flowering	0.5994	0.6307	0.6362
fruiting	0.5059	0.5181	0.4998
seeding	0.6548	**0.7755**	**0.8031**

Fruiting is the weakest class for every segmenter — a task/data property (smallest class, fruit occludes and fragments the canopy silhouette) rather than something specific to either architecture.

4.3 End-to-end pipeline results

Pipeline accuracy and macro metrics by segmenter variant

SEGMENTER	PIPELINE ACCURACY	MACRO PRECISION	MACRO RECALL	MACRO F1
DeepLabV3-ResNet50	84.38%	0.83	0.84	0.83
U-Net (native resize)	78.75%	0.80	0.83	0.78
U-Net (crop-aligned)	**89.38%**	**0.87**	**0.88**	**0.88**

Confusion matrix — U-Net, native resize (rows = true, columns = predicted)

	DEVELOPING	FLOWERING	FRUITING	SEEDING
developing	**39**	2	0	0
flowering	2	**31**	27	0
fruiting	0	2	**18**	1
seeding	0	0	0	**38**

Confusion matrix — U-Net, crop-aligned (rows = true, columns = predicted)

	DEVELOPING	FLOWERING	FRUITING	SEEDING
developing	**39**	2	0	0
flowering	2	**51**	7	0
fruiting	0	6	**15**	0
seeding	0	0	0	**38**

WHAT CROP-ALIGNMENT FIXED

With native-resize preprocessing, 27 of 60 flowering images were misclassified as fruiting — the geometry mismatch between U-Net's mask and the classifier's crop was silently feeding the classifier a misaligned image. Re-running U-Net on the classifier's own crop geometry cut that to 7 of 60 and lifted pipeline accuracy from 78.75% to **89.38% (+10.6 pts)**, while segmentation IoU stayed essentially flat (0.6646 → 0.6674) — the fix was almost pure upside, confirming the drop was a pipeline-alignment artifact, not a segmentation-quality problem.

5. Cross-model comparison

All three segmenters, same pipeline, same 160-image test set

SEGMENTER	FOREGROUND IOU	FOREGROUND DICE	MEAN IOU	PIPELINE ACCURACY
DeepLabV3-ResNet50	0.6230	0.7565	0.7507	84.38%
U-Net (native resize)	0.6646	0.7844	**0.7926**	78.75%
U-Net (crop-aligned)	**0.6674**	0.7827	0.7753	**89.38%**

Taken in isolation, U-Net segments plant pixels slightly more accurately than DeepLabV3 on every IoU/Dice measure. But segmentation quality alone does not predict pipeline quality: the native-resize U-Net pipeline is the *worst* performer of the three (78.75%) purely because of a preprocessing-geometry mismatch with the

downstream classifier, not because its masks were worse. Once that mismatch is removed, the crop-aligned U-Net pipeline becomes the best of the three (89.38%), 5 points above DeepLabV3. The practical conclusion for deployment: **segmenter choice matters less than making sure the segmenter's preprocessing convention matches the classifier's** — an architecturally "better" segmenter can still produce a worse end-to-end system if it is bolted on without checking that alignment.

6. Challenges encountered

- **Background dependency in the naive classifier.** The original ResNet50 loses 20 accuracy points when background is removed at test time, and stops predicting "fruiting" entirely — motivating the whole masked/segmentation line of work (§2.3).
- **Masked training doesn't generalize on its own.** A classifier trained purely on artificially masked images collapses to 35.63% on real photos, having learned features tied to the black background and mask edges rather than the plant — this is why the final pipeline needs a *predicted*-mask stage rather than simply shipping the masked classifier (§2.3).
- **Grad-CAM and accuracy disagree on "who relies on background more."** The accuracy drop says the original model is more background-dependent; the Grad-CAM attention-fraction metric shows both models spending >70% of activation mass outside the plant polygon, with no significant difference between them (§2.4). Flagged as an open tension rather than resolved — it suggests accuracy-based and attribution-based measures of "reliance" are capturing different things and shouldn't be assumed to agree.
- **Joint image/mask augmentation.** A naive `transforms.Compose` applied separately to an image and its mask silently applies different random parameters to each, misaligning them. Fixed by switching to `torchvision.transforms.v2` with paired `tv_tensors` so identical random crop/flip/rotation parameters are shared between image and mask.
- **Memory blowout during evaluation.** Materializing full-resolution prediction/target arrays per image during segmentation scoring caused a `MemoryError`. Fixed by accumulating a running pixel-level confusion matrix batch-by-batch instead of holding every image's arrays in memory at once.
- **Checkpoint / state-dict mismatch on export.** Training used DeepLabV3's default weights, which auto-attach an auxiliary classifier head; the deployed inference model is built without one. Auxiliary-head weights had to be stripped before `load_state_dict` to avoid a key-mismatch error.
- **Geometry mismatch between segmenters and the classifier (U-Net).** U-Net's native direct-resize convention silently misaligned masks with the classifier's `Resize→CenterCrop` convention, costing 10.6 points of pipeline accuracy until a crop-aligned variant was introduced (§4.1, §4.3).
- **Oracle-mask vs. predicted-mask gap.** Ground-truth polygon masks aren't available for new, unannotated photos. The realistic deployable pipeline (predicted masks) underperforms the oracle-mask ceiling by 4.37 points (91.87% → 87.50%) — a gap that has to be reported as the real-world number, not the oracle one (§3.4).
- **Fixed-split accuracy overstates real performance.** The single held-out test split's DeepLabV3 pipeline accuracy (87.50%) sits 5.3 points above the 5-fold cross-validated mean (82.23% ± 2.16), showing the fixed split was somewhat favorable — the CV mean is the more defensible figure to lead with (§3.5).
- **Class imbalance.** Fruiting (21 test images, roughly a third the size of flowering) is the weakest class in every single experiment in this study — every classifier and every segmenter — indicating this is a

data/task property (visual similarity to flowering, occlusion by fruit) rather than a fixable modeling choice within the current dataset.

- **Small unexplained run-to-run drift.** A later standalone re-evaluation of the DeepLabV3 pipeline reproduced 84.38% accuracy / 0.7507 mean IoU against an earlier "originally reported" 85.00% / 0.7570 from the same checkpoint — a sub-1-point discrepancy the notebooks note but don't diagnose, most likely non-determinism in batching or mask thresholding rather than a real difference in the model.

7. Summary of key findings

1. A plain ResNet50 classifier is measurably background-dependent: removing background at test time costs 20 accuracy points and zeroes out fruiting-stage recall entirely.
2. Training on masked images alone is not a fix — it trades one shortcut (background) for another (mask-edge artifacts) and collapses to 35.63% accuracy on real photos.
3. A segment→mask→classify pipeline with a *predicted* mask is a workable middle ground: DeepLabV3-ResNet50 reaches 84.38–87.50% pipeline accuracy (82.23% ± 2.16 under 5-fold CV) with no ground-truth annotation needed at inference time, a 4.37-point gap below the 91.87% oracle-mask ceiling.
4. U-Net segments plant pixels marginally more accurately than DeepLabV3 in isolation (mean IoU 0.79 vs. 0.75), but that advantage is easily erased or reversed by preprocessing mismatches with the downstream classifier — the crop-aligned U-Net pipeline is the best system tested overall (89.38%), while the naively-integrated version is the worst (78.75%).
5. Fruiting is the hardest growth stage across every model and every experiment — a genuine data-property limitation (smallest class, occlusion, visual overlap with flowering) rather than an architecture choice to fix.
6. Attribution (Grad-CAM) and accuracy-based measures of "background reliance" disagree, which is itself a finding worth reporting: the accuracy story is cleaner than the attention-map story, and the thesis should not claim more agreement between the two than the numbers show.

Compiled from `resnet50.ipynb`, `resnet50_masked_comparison.ipynb`, `leaf_segmentation.ipynb`, `unet_leaf_pipeline_comparison.ipynb`, and `deeplabv3_pipeline_evaluation.ipynb`, with figures cross-checked against `masked_vs_original_comparison.csv`, `unet_vs_deeplabv3_comparison_v2.csv/json`, and `deeplabv3_pipeline_kfold_cv_results.csv`. All numbers reproduced verbatim from cell outputs and saved result files — none re-derived or estimated.